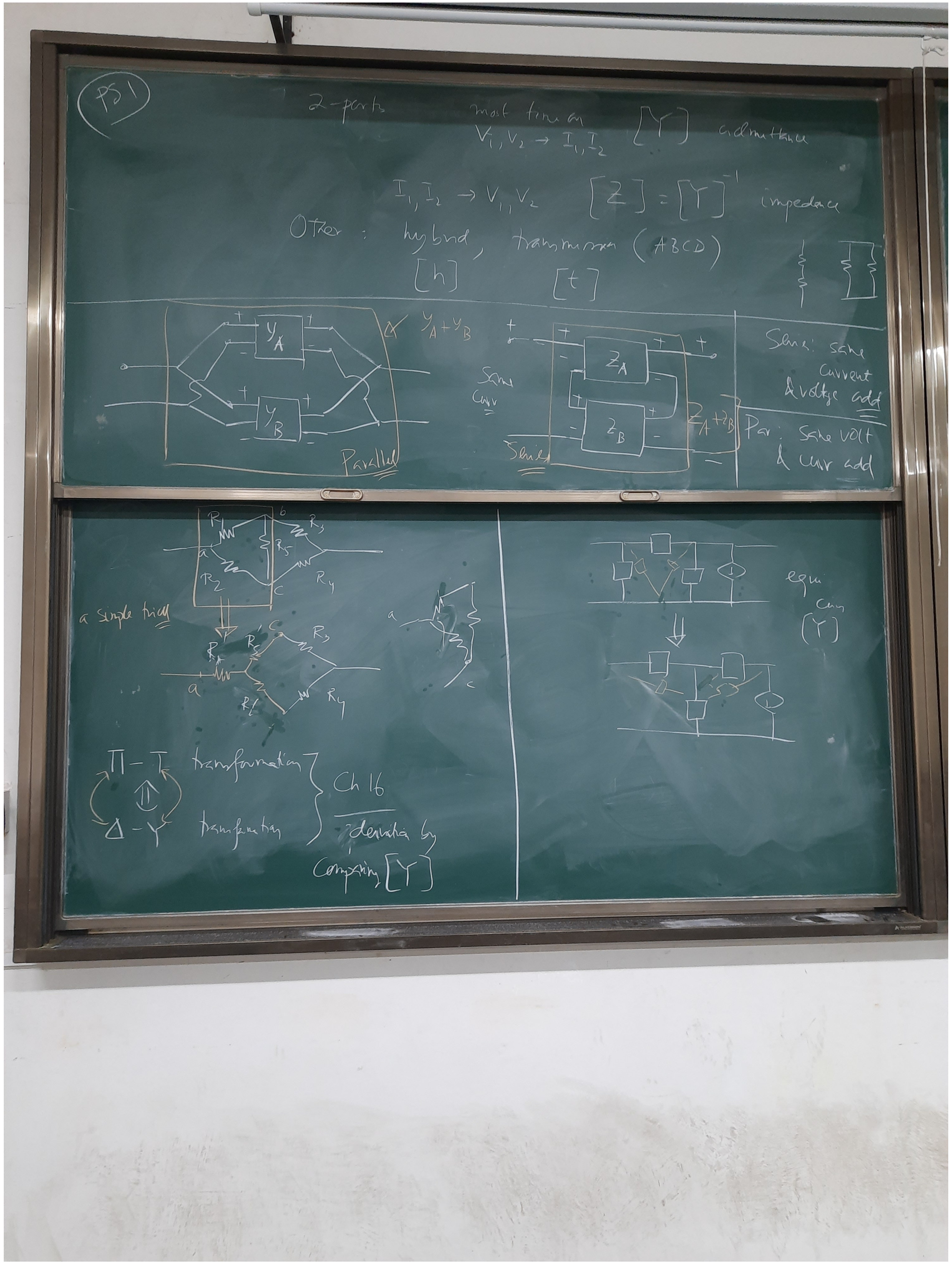


PS1

2-ports most time on $V_1, V_2 \rightarrow I_1, I_2$ $[Y]$ admittance

$I_1, I_2 \rightarrow V_1, V_2$ $[Z] = [Y]^{-1}$ impedance

Other : hybrid, transmission (ABCD)

$[h]$ $[t]$

$y_A + y_B$

Same curr

$Z_A + Z_B$

Series: same current & voltage add

Par: same volt & curr add

a simple trick

$\Pi - T$ transformation

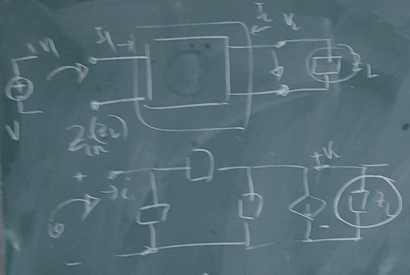
$\Delta - Y$ transformation

Ch 16

derivation by comparing $[Y]$

equ circ (Y)

$= \frac{1}{Z_{22}}$

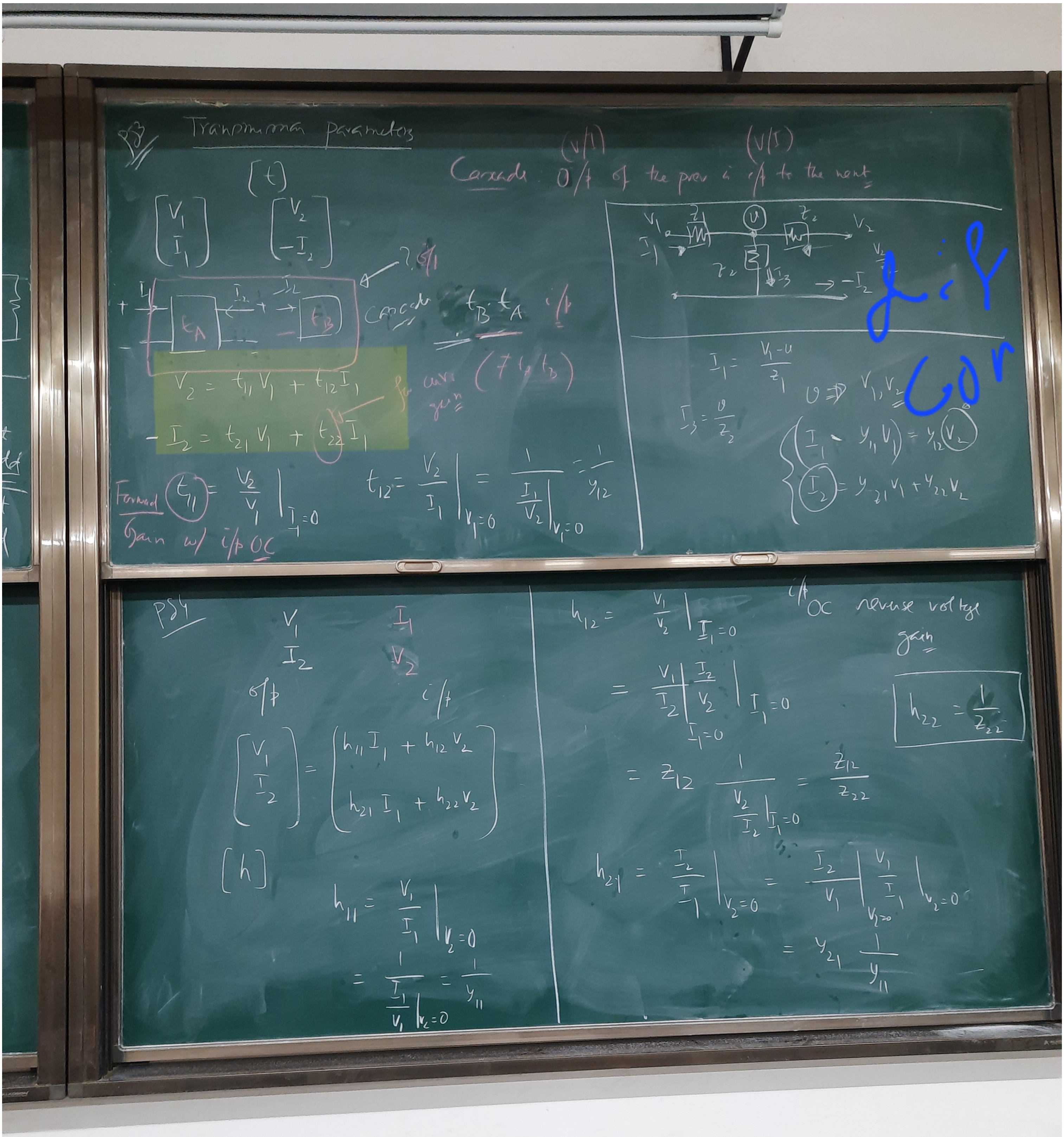


App^r 1 $\quad Z_{in}(Z_L) = \frac{v}{i}$

App^r 2 Simpler

$V_1 = Z_{11} I_1 + Z_{12} I_2$

$-Z_L I_2 \; \boxed{V_2} = Z_{21} I_1 + Z_{22} I_2 \ldots$



Transmission parameters

$[t]$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} \qquad \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

? S/I

Cascade : O/P of the prev is i/p to the next

(V/I) (V/I)

Cascade $t_B \cdot t_A$ i/p

$(\neq t_A t_B)$

$$V_2 = t_{11} V_1 + t_{12} I_1$$

$$-I_2 = t_{21} V_1 + t_{22} I_1$$

for curr gain

Forward $t_{11} = \left.\frac{V_2}{V_1}\right|_{I_1=0}$

Gain w/ i/p OC

$$t_{12} = \left.\frac{V_2}{I_1}\right|_{V_1=0} = \frac{1}{\left.\frac{I_1}{V_2}\right|_{V_1=0}} = \frac{1}{y_{12}}$$

$$I_1 = \frac{V_1 - v}{Z_1}$$

$$I_3 = \frac{v}{Z_2}$$

$v \Rightarrow V_1, V_2$

$$\begin{cases} \left(I_1 - y_{11} V_1\right) = y_{12} V_2 \\ I_2 = y_{21} V_1 + y_{22} V_2 \end{cases}$$

PSY

V_1 I_1

I_2 V_2

o/p i/p

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} I_1 + h_{12} V_2 \\ h_{21} I_1 + h_{22} V_2 \end{bmatrix}$$

$[h]$

$$h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0} = \frac{1}{\left.\frac{I_1}{V_1}\right|_{V_2=0}} = \frac{1}{y_{11}}$$

$$h_{12} = \left.\frac{V_1}{V_2}\right|_{I_1=0}$$

i/p OC reverse voltage gain

$$= \left.\frac{V_1}{I_2}\right|_{I_1=0} \left.\frac{I_2}{V_2}\right|_{I_1=0}$$

$$= Z_{12} \frac{1}{\left.\frac{V_2}{I_2}\right|_{I_1=0}} = \frac{Z_{12}}{Z_{22}}$$

$$\boxed{h_{22} = \frac{1}{Z_{22}}}$$

$$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0} = \left.\frac{I_2}{V_1}\right|_{V_2=0} \left.\frac{V_1}{I_1}\right|_{V_2=0}$$

$$= y_{21} \frac{1}{y_{11}}$$